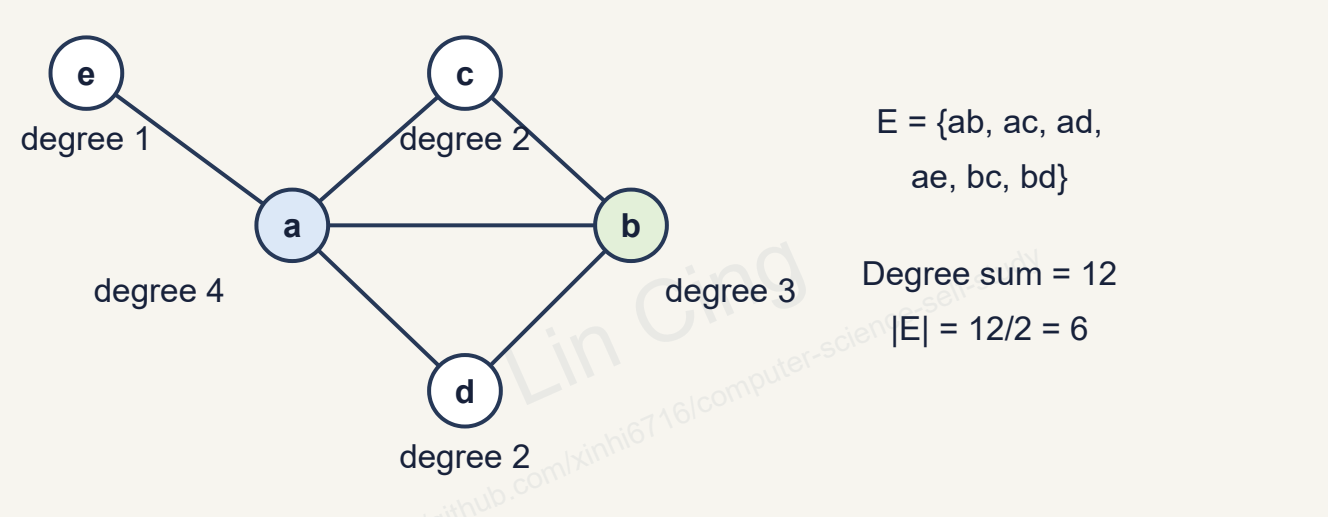
Graph-theory solutions

1

The degree sum is twice the edge count and thus even. Even-degree vertices contribute an even sum, so odd-degree vertices also have an even sum. A sum of s odd integers has parity s; therefore s is even.

2

The sum is twelve, so six edges are required. Havel–Hakimi first connects degree-four vertex to all four others, leaving sorted residual degrees (2,1,1,0). Connect degree two to the two positive degrees, leaving zeros. Explicit vertices a, b, c, d, e and edges ab, ac, ad, ae, bc, bd give degrees 4,3,2,2,1.



3

The sum is 2・3+6・4=30=2e, giving e=15. Connectivity and exactly two odd vertices guarantee an Euler trail with those degree-three endpoints. There is no Euler circuit.

4

$K_{3,3}$ is simple bipartite and has no cycle shorter than four. A planar embedding would therefore satisfy $4f \leq 2e$, counting each edge side once. It is connected, so Euler's formula with v=6,e=9 gives f=5. But 4f=20>18=2e, contradiction. This excludes every embedding.

Graph-theory solutions (continued)

5

Name cyclic vertices v_1,v_2,v_3,v_4 and split by opposite v_1,v_3 colors. Equal colors give k choices for that color and k−1 choices independently for each of v_2,v_4: $k(k-1)^2$. Distinct colors give $k(k-1)$ ordered choices and $(k-2)^2$ for the others. Hence

$$P_{C_4}(k) = k(k-1)^2 + k(k-1)(k-2)^2 = k(k-1)(k^2-3k+3).$$

At k=2 it gives two alternating colorings, as expected for this connected bipartite graph.

6

Deleting a bridge separates two sides. A cycle using it would need another edge crossing back between the sides, contradicting that it is the unique cut edge. Thus no Hamilton cycle uses a bridge.

After connecting two K_5 graphs, n=10. Bridge endpoints have degree five, but the other eight vertices have degree four<5 and fail Dirac's universal threshold. In each clique choose a Hamilton path ending at its bridge endpoint; concatenate across the bridge to visit all ten vertices once. A Hamilton cycle would need to cross between cliques twice, impossible with the only crossing edge being that bridge. This is not a counterexample to Dirac; its degree premise fails.

7

Both have V={A, B, C, D}, E={AB, AC, AD, BC, BD, CD}. The crossing of AC, BD is not in V and does not split either edge, so there remain four vertices and six edges. The crossed drawing alone proves nothing about nonplanarity, while the right crossing-free drawing proves K_4 planar.

Initially (v, e, f)=(4,6,4). AD is not a bridge because A–B–D remains; deleting it merges F_1,F_3, giving (4,5,3). BD is then not a bridge because B–C–D remains; deletion merges inner faces, giving (4,4,2). AC still has alternate A–B–C, so deletion gives (4,3,1) by merging inner and outer faces. Each step preserves v−e+f=2. Remaining AB, BC, CD form a tree. CD is now a bridge; deleting it would isolate D and cannot use the two-face merge argument.

For $K_3 \sqcup K_2$, v=5,e=4,c=2. The triangle encloses one face and all components share the unbounded face, so f=2. Then 5−4+2=3=1+c. Separate outer-face counts would duplicate one geometric region.

Graph-theory solutions (continued)

8

In A, B, C, D order both outdegrees and indegrees are 2,1,2,1. The nonisolated underlying graph is connected and every vertex balanced. A circuit is A→B→C→A→C→D→A, using AB, BC, CA, AC, CD, DA once each. There is no open Euler trail with different endpoints, since no vertices have imbalances +1,−1.

Adding X→Y→Z→X leaves old degrees unchanged and gives each new vertex indegree/outdegree one. Balance remains but nonisolated underlying connectivity fails: two components cannot be covered by one trail. All nine arcs cannot share a circuit. Both degree and connectivity conditions are essential.

9

There are 6 • 7=42 vertices. Each has four distinct neighbors (i±1,j),(i, j±1), giving degree four and e=42 • 4/2=84.

Every step changes exactly one coordinate. In a three-step closed walk changing both coordinates, one coordinate changes exactly once; ±1 cannot vanish modulo six or seven. If all three steps change one coordinate, the total change belongs to {−3,−1,1,3}, again nonzero modulo either. Thus no triangle exists. The graph is simple and connected, since coordinates can be adjusted consecutively to reach any vertex.

A triangle-free planar graph would have $e \le 2v-4=80$, contradicting 84. The graph actually has a seven-cycle in the second-coordinate direction and is not bipartite. The proof uses triangle-freeness, not a false claim that every cycle is even.

10

Five vertices have ten possible edges, so exactly four gives $\binom{10}{4} = 210$. Inclusion–exclusion for no isolates gives

$$2^{10} - 5 \cdot 2^6 + 10 \cdot 2^3 - 10 \cdot 2 + 5 - 1 = 1024 - 320 + 80 - 20 + 5 - 1 = 768.$$

For connectivity use c_1=1,c_2=1,c_3=4,c_4=38 and classify the component of vertex one. Disconnected contributions at k=1,2,3,4 are

$$\binom{4}{0} \cdot 1 \cdot 2^6 = 64, \quad \binom{4}{1} \cdot 1 \cdot 2^3 = 32, \quad \binom{4}{2} \cdot 4 \cdot 2 = 48, \quad \binom{4}{3} \cdot 38 = 152.$$

Thus c_5=1024−(64+32+48+152)=728. The difference forty consists of no-isolate disconnected graphs with component sizes 2+3. Choose the unique two-vertex component in ten ways; it has one edge, and the remaining three vertices have four connected graphs, giving 10 • 4=40. No division by two is needed because the different component sizes distinguish them uniquely.

Graph-theory solutions (continued)

11

Removing |S| vertices from a Hamilton path leaves at most |S|+1 path segments. Other edges in G−S can only merge them, proving c(G−S)≤|S|+1, including empty S, which requires original connectivity.

For $K_{2,4}$ delete the two-vertex part. Four isolates remain, violating 4≤3, so no Hamilton path exists. Equivalently an alternating path cannot use part sizes differing by more than one.

$K_{3,4}$ has path $b_1–a_1–b_2–a_2–b_3–a_3–b_4$. Every edge crosses parts and each vertex occurs once. A cycle would alternate equally many vertices from both parts, impossible for 3≠4. Alternatively deleting the three-vertex part leaves four components, violating the Hamilton-cycle bound.

12

Sequential coloring gives P_C_3(k)=k(k−1)(k−2). Deleting an edge of C_4 gives a four-vertex tree; contracting it gives C_3:

$$P_{C_4} = k(k-1)^3 - k(k-1)(k-2) = k^4 - 4k^3 + 6k^2 - 3k.$$

For C_5 deletion gives a five-vertex tree and contraction C_4:

$$P_{C_5} = k(k-1)^4 - P_{C_4} = k^5 - 5k^4 + 10k^3 - 10k^2 + 4k.$$

At two, 32−80+80−40+8=0: an odd cycle is not two-colorable. At three, 243−405+270−90+12=30 proper colorings with three named colors exist. Every one actually uses all three, since using fewer would give a two-coloring. Thus $\chi(C_5)$=3; thirty counts colorings, not the minimum number of colors.

13

Each adjacency list stores IDs in order: A:[1,3], B:[1,2], C:[2,3,4,6], D:[4,5], E:[5,6]. Initially S=[A], answer=[]. The complete trace is:

Graph-theory solutions (continued)

Action	S (top at right)	answer (append order)
e1	[A,B]	[]
e2	[A,B,C]	[]
e3	[A,B,C,A]	[]
pop A	[A,B,C]	[A]
e4	[A,B,C,D]	[A]
e5	[A,B,C,D,E]	[A]
e6	[A,B,C,D,E,C]	[A]
pop C	[A,B,C,D,E]	[A,C]
pop E	[A,B,C,D]	[A,C,E]
pop D	[A,B,C]	[A,C,E,D]
pop C	[A,B]	[A,C,E,D,C]
pop B	[A]	[A,C,E,D,C,B]
pop A	[]	[A,C,E,D,C,B,A]

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The first pop removes only the final A frame. C remains on the stack and has unused edge e4, so the search continues. Reversing the final answer gives A–B–C–D–E–C–A and IDs 1,2,4,5,6,3. Each ID occurs once, including e3 at the end.

After replacing EC by EF, degrees are A:2, B:2, C:3, D:2, E:2, F:1. The nonisolated part is connected, so an open Euler trail exists, but its endpoints must be C and F. Starting at A violates the start condition. With that start, the algorithm produces A–B–C–D–E–F–A after reversal: the final pair F–A is not an edge. A count of m used IDs and m+1 output vertices alone does not guarantee a valid route when the preconditions are omitted. Starting at C instead gives C–B–A–C–D–E–F with IDs 2,1,3,4,5,6.

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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